

Logos Experiment 20: Efficiency Index and Resource-Aware Ranking

Objective

Experiment 20 converts the Logos experiment series from raw measurements into a resource-aware evaluation framework. No model is trained in this experiment. It ranks completed models by quality and cost using validation loss, perplexity, parameters, checkpoint size, VRAM, latency, and energy.

Why this matters

The main project goal is not only to reduce perplexity, but to improve capability per parameter, memory byte, latency unit, checkpoint size, and watt. A single loss table is therefore incomplete. Experiment 20 introduces a ranking layer that separates pure quality, deployment efficiency, compactness, and scaling behavior.

Score definitions

Metric	Definition	Interpretation
Quality score	$\text{best_ppl} / \text{model_ppl}$	Higher means closer to best observed perplexity.
Checkpoint score	$\text{smallest_checkpoint} / \text{model_checkpoint}$	Higher means smaller saved model.
VRAM score	$\text{lowest_vram} / \text{model_vram}$	Higher means lower GPU memory use.
Latency score	$\text{fastest_latency} / \text{model_latency}$	Higher means faster generation.
Energy score	$\text{lowest_energy} / \text{model_energy}$	Higher means lower training energy.
Parameter score	$\text{smallest_params} / \text{model_params}$	Higher means fewer parameters.
Deployment LEI	$0.45 \text{ quality} + 0.20 \text{ checkpoint} + 0.15 \text{ VRAM} + 0.15 \text{ latency} + 0.05 \text{ parameters}$	Inference/deployment-oriented score.
Training-aware LEI	$0.40 \text{ quality} + 0.15 \text{ checkpoint} + 0.10 \text{ VRAM} + 0.10 \text{ latency} + 0.10 \text{ parameters} + 0.15 \text{ energy}$	Training-cost-aware score.

Important caution

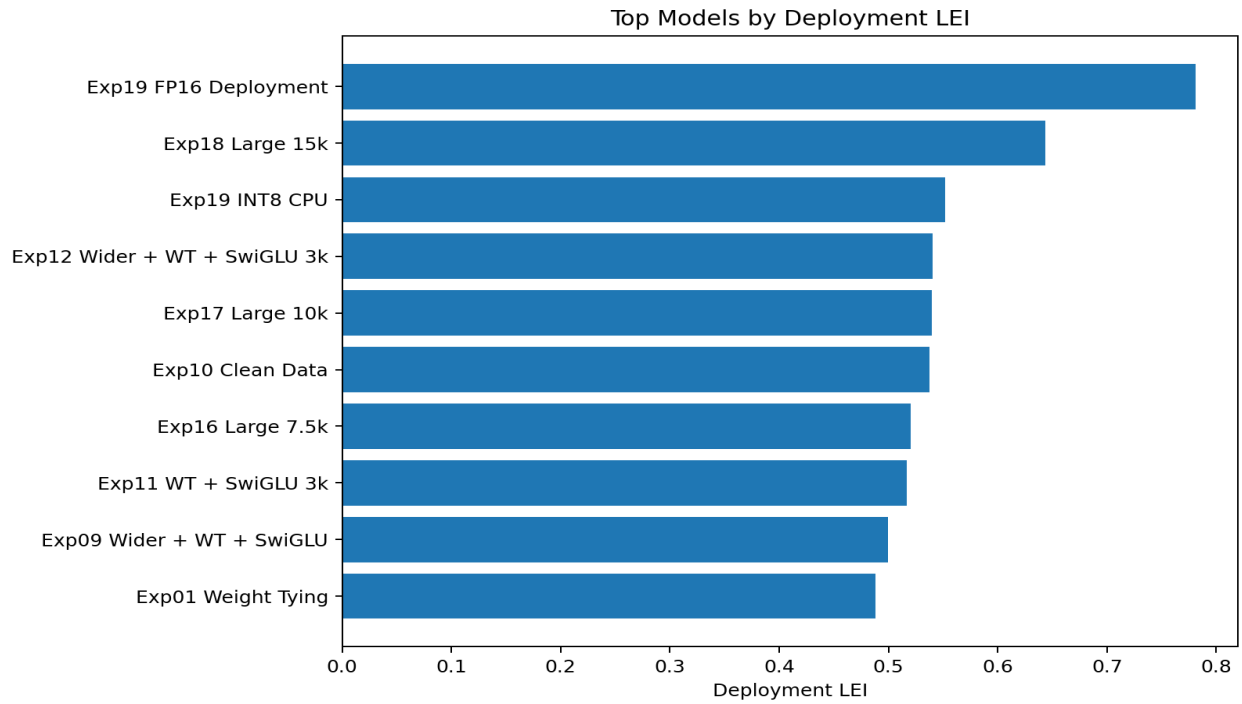
LEI is a comparative ranking tool, not a universal law. The weights are intentionally explicit and can be changed for different deployment priorities. Compression results from Experiment 19 use sampled validation estimates, so small quality differences between FP32, FP16, and INT8 should be treated as evaluation noise.

Key ranking results

Category	Best result	Reason
Best trained quality	Exp18 Large 15k	Best final trained model before compression: PPL 53.57, val loss 3.9811.
Best deployment model	Exp19 FP16 Deployment	Good estimated quality, 310.24 MB checkpoint, 5.96 GB peak VRAM, 9.14 ms/token.
Best compact small model	Exp12 Wider + WT + SwiGLU 3k	18.54M params, 70.76 MB checkpoint, PPL 210.00.
Best CPU compressed model	Exp19 INT8 CPU	Smallest state at 267.05 MB, but slower at 20.34 ms/token.

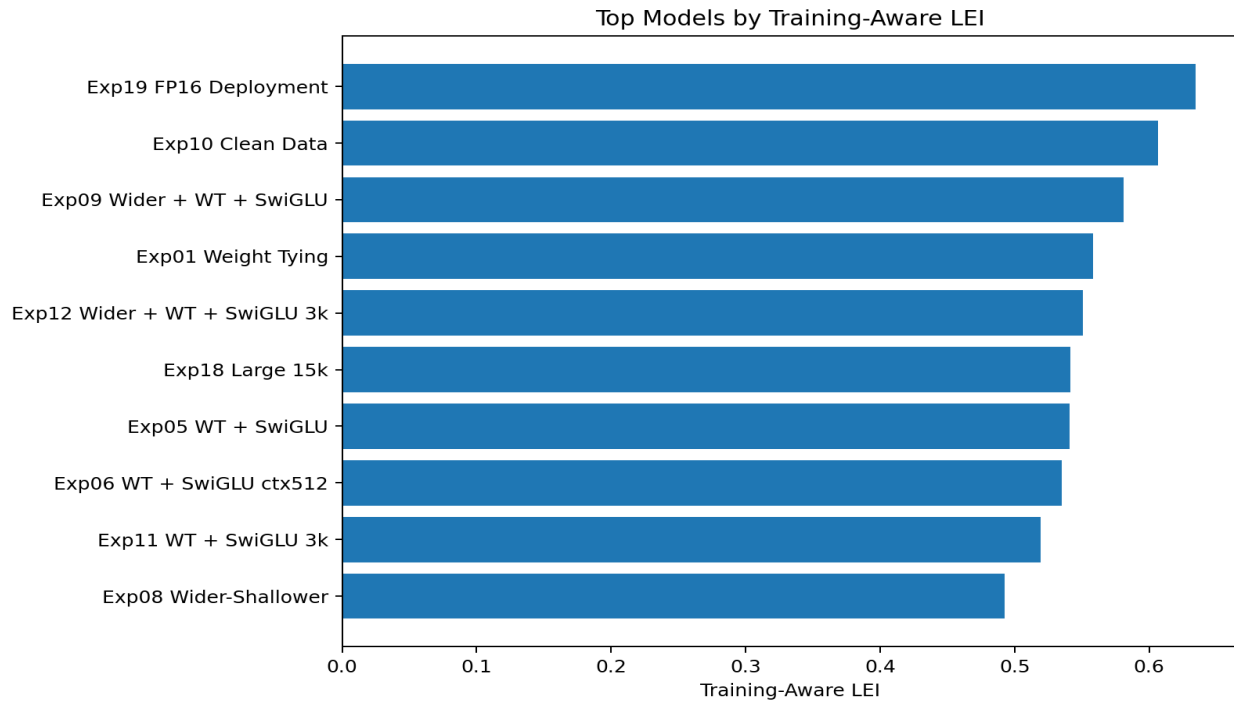
Top 10 by Deployment LEI

Experiment	Category	PPL	Checkpoint MB	VRAM GB	ms/token	Deploy LEI
Exp19 FP16 Deployment	compression	50.89	310.24	5.96	9.14	0.7808
Exp18 Large 15k	scaling	53.57	473.20	17.73	9.11	0.6434
Exp19 INT8 CPU	compression	52.77	267.05	N/A	20.34	0.5518
Exp12 Wider + WT + SwiGLU 3k	longer_train	210.00	70.76	10.54	12.06	0.5406
Exp17 Large 10k	scaling	61.36	473.20	17.73	14.77	0.5400
Exp10 Clean Data	data_test	338.53	70.76	10.54	8.65	0.5374
Exp16 Large 7.5k	scaling	68.40	473.20	17.73	11.92	0.5205
Exp11 WT + SwiGLU 3k	longer_train	202.82	73.45	11.18	14.03	0.5169
Exp09 Wider + WT + SwiGLU	small_efficiency	313.87	70.76	10.54	12.71	0.4996
Exp01 Weight Tying	small_ablation	320.32	73.44	10.69	12.64	0.4883



Top 10 by Training-Aware LEI

Experiment	Category	PPL	Checkpoint MB	ms/token	Wh	Train LEI
Exp19 FP16 Deployment	compression	50.89	310.24	9.14	N/A	0.6343
Exp10 Clean Data	data_test	338.53	70.76	8.65	8.98	0.6066
Exp09 Wider + WT + SwiGLU	small_efficiency	313.87	70.76	12.71	9.06	0.5813
Exp01 Weight Tying	small_ablation	320.32	73.44	12.64	9.87	0.5582
Exp12 Wider + WT + SwiGLU 3k	longer_train	210.00	70.76	12.06	16.29	0.5507
Exp18 Large 15k	scaling	53.57	473.20	9.11	277.39	0.5412
Exp05 WT + SwiGLU	small_combined	306.36	73.44	15.04	10.49	0.5408
Exp06 WT + SwiGLU ctx512	context_test	345.43	73.69	13.62	10.72	0.5351
Exp11 WT + SwiGLU 3k	longer_train	202.82	73.45	14.03	19.48	0.5191
Exp08 Wider-Shallower	shape_test	318.00	126.15	8.53	9.63	0.4924



Interpretation

The safest interpretation is to keep multiple leaderboards. A single aggregate LEI can over-reward very cheap but weak models or over-reward high-quality compressed models when validation is sampled. The corrected reporting strategy is therefore: Exp18 is the best trained model, Exp19 FP16 is the best deployment version, Exp12 is the best compact small model, and Exp13-Exp18 form the main scaling evidence.

Generated outputs

This experiment produces a CSV file with all normalized scores, a JSON summary, and ranking plots for deployment efficiency, training-aware efficiency, and scaling behavior.